BENCH TEST REPORT · SAVED RECORD · OFFLINE REVIEW (HISTORICAL)

HMI operator trial (Muse-operated browser HMI)

CAMPAIGN OBJECTIVE (MUSE PLAN) Prove actual HMI operation on same controlled browser: operate Auto/Start/Stop/Reset via HMI, verify pixels vs PLC outputs, induce/clear declared fault for Reset and alarm/mode, assess UI-01 to UI-04, finish Stopped fault-free no active item/completion.

Live PLC evidence | Run completed | Historical record | `portal-20261103232137992-1`

Summary

Generated from the campaign record; no additional model call

- Run completed 2026-11-03 after 1 min 53 s.
- 3 of 3 planned experiments reached a recorded conclusion (2 observed, 1 inconclusive).
- Muse recorded 4 requirement assessments: 3 pass, 0 fail, 1 inconclusive; none reviewed in the campaign record.
- 1 finding with 10 exact PLC claims; 1 unreviewed.
- Separate review files attached: s2-audit.json (reference review required); engineering-review.json (pass with limitations).

CAMPAIGN EXECUTION

Recorded run status

Run completed

EXPERIMENTS

Experiment conclusions as recorded; observed is not pass

3	Planned
2	Observed
1	Inconclusive
0	Not run

REQUIREMENT VERDICTS

Muse-recorded assessments; unreviewed unless stated

3	Pass
0	Fail
1	Inconclusive
0	No verdict

FINDINGS

Engineer review state in the campaign record

1	Recorded
1	Unreviewed
0	Accepted

MUSE ACTIONS

Tool calls, counted by outcome

29	Applied
0	Refused
0	Timed out

Needs attention

> EXP-HMI-03: inconclusive (UI-04).

> UI-04 assessed inconclusive by Muse (ASSESS-UI04-01).

> ASSESS-UI03-01 (UI-03): Cites #2783: pixels were not among the images sent to Muse (model exchange log); also flagged by s2-audit.json and engineering-review.json.

> 1 cited image not available in this report; placeholders keep their references.

> 1 finding awaiting engineer review in the campaign record.

Execution

Completed 2026-11-03 23:23:43.443 UTC.

Muse summary MODEL-AUTHORED · UNREVIEWED

> UI checkpoint completed via actual HMI clicks on controlled browser. UI-01 pass: START/STOP/RESET/AUTO labels with Applied-by-PLC or Readback-confirmed visible results corroborated by PLC acks. UI-02 pass: HMI separately displayed motor/divert command vs observed feedback, including commanded-no-feedback and off-while-running states. UI-03 pass: injected drive overload showed Faulted mode and 'Drive overload' fault text on Operation, matching PLC FaultCode 10/FaultText. UI-04 inconclusive: only Fresh data observed, no stale/unknown or WCS disconnect was induced. Reset after clearing overload returned PLC to Stopped, FaultCode 0, Motor_RunCmd false, ItemActive false, CompletionPending false; Auto turned off. Finding FIND-HMI-01 records exact PLC claims with injected truth kept separate.

Identity and source

Live PLC evidence

Campaign	`portal-20261103232137992-1`
Report title	HMI operator trial (Muse-operated browser HMI)
Machine	*Name not recorded in the campaign record; identified by the specification and I/O map below*
Evidence class	Live PLC evidence
Report source	Saved record · offline review (historical)
Started	2026-11-03 23:21:50.053 UTC
Ended	2026-11-03 23:23:43.443 UTC
Duration	1 min 53 s
Controller build	`8001/interface-1/sha256:6c2a1eb6c9c83485b75cb9bcefc1be237a94c8727ecf8098ad57a4d680516628`
World module	`sha256:277992048f8ca9a8b36f2f90c0d4c33ef479cb8b3c355d6a2233a272850698f5;worldVersion:0.5.0;loaded-by-child:53648`
Specification	`sha256:acd1e39619af301155ac0f8b81f4b824c36c478800b4d482ddb225a2a21d5185`
I/O map	`f8c30539e336f55056f73d4d7ef467b10a9175a60dec3ec156ba1a7bb7a7ba80`
Model	`muse-wonder-1.0`

Connector	`m8-opcua-v1/m8-conveyor-1.0.2;controlsbench.wonder-process.v1;child:b7dfb3a1d66f2722`
PLC boot identity	`1787886077538`
Authorized scope	normal, faults · known diagnosis

Claim boundary: Wonder-directed M8 campaign through the runtime owner's verified isolated adapter. Scope-limited experiment evidence; no certified controller coverage. HMI frames remain separate.

Test scope and results

3 experiments

Experiment	Requirement refs	Recorded conclusion	Requirement verdict	Evidence
`EXP-HMI-01` Planned	UI-01, UI-02	Observed — conclusion recorded, not a pass verdict	UI-01: pass (Muse assessment, unreviewed); UI-02: pass (Muse assessment, unreviewed)	`#798 HMI action #938 HMI action #952 Observation #1120 HMI action #1131 Observation #1197 PLC output #1238 PLC output #1255 HMI frame #1257 HMI frame`
`EXP-HMI-02` Planned	UI-01, UI-03, UI-02	Observed — conclusion recorded, not a pass verdict	UI-01: pass (Muse assessment, unreviewed); UI-02: pass (Muse assessment, unreviewed); UI-03: pass (Muse assessment, unreviewed)	`#2508 HMI action #2653 Observation #2745 PLC output #2783 HMI frame #2798 HMI frame #3722 HMI action #3733 Observation #3980 PLC output #4015 PLC output #4075 HMI frame`
`EXP-HMI-03` Planned	UI-04	Inconclusive	UI-04: inconclusive (Muse assessment, unreviewed)	`#5931 Assessment #4075 HMI frame #2798 HMI frame #4015 PLC output #6041 HMI frame`

▾ `EXP-HMI-01` normal Observed — conclusion recorded, not a pass verdict

UI-01/UI-02: Use HMI Auto On, Start, Stop; verify visible controls/state and PLC Mode/Motor_RunCmd independently.

Objective	MUSE UI-01/UI-02: Use HMI Auto On, Start, Stop; verify visible controls/state and PLC Mode/Motor_RunCmd independently.
Requirement refs	UI-01, UI-02
Origin	Planned
Run	2026-11-03 23:22:02.965 UTC → 2026-11-03 23:22:29.922 UTC

Stimulus / operator action	• HMI action cmd.auto.on — applied `#798 HMI action` • HMI action cmd.start — applied `#938 HMI action` • HMI action cmd.stop — applied `#1120 HMI action`
Expected behaviour	*No explicit acceptance criterion is recorded in the campaign for this experiment.*
Observed behaviour	• Wait for Motor_RunCmd = true (up to 5000 ms) — observed `#952 Observation #947 PLC output` • Wait for Motor_RunCmd = false (up to 5000 ms) — observed `#1131 Observation #1129 PLC output` Cited PLC output `#1197 PLC output` recorded 2026-11-03 23:22:08.003 UTC (source 2026-11-03 23:22:07.979 UTC) · PLC cycle 34122 `Motor_RunCmd false` Cited PLC output `#1238 PLC output` recorded 2026-11-03 23:22:08.584 UTC (source 2026-11-03 23:22:08.556 UTC) · PLC cycle 34150 `Motor_RunCmd false`
Recorded conclusion	Observed — conclusion recorded, not a pass verdict
Requirement verdict	UI-01: pass (Muse assessment, unreviewed); UI-02: pass (Muse assessment, unreviewed)
Supporting evidence	`#798 HMI action #938 HMI action #952 Observation #1120 HMI action #1131 Observation #1197 PLC output #1238 PLC output #1255 HMI frame #1257 HMI frame`
Limitations	None recorded.

▾ **EXP-HMI-02** faults Observed — conclusion recorded, not a pass verdict

UI-01/UI-03/UI-02: Induce declared drive overload fault, verify HMI fault text/mode and PLC fault/motor, clear fault then HMI Reset, verify no auto-restart.

Objective	MUSE UI-01/UI-03/UI-02: Induce declared drive overload fault, verify HMI fault text/mode and PLC fault/motor, clear fault then HMI Reset, verify no auto-restart.
Requirement refs	UI-01, UI-03, UI-02
Origin	Planned
Run	2026-11-03 23:22:29.930 UTC → 2026-11-03 23:23:27.261 UTC
Stimulus / operator action	• HMI action cmd.start — applied `#2508 HMI action` • Inject fault.setOverload {"active":true} — applied `#2627 Source record #2628 Raw inputs #2629 PLC ack #2630 PLC output` • Clear fault.setOverload {"active":false} — applied `#3671 Source record #3672 Raw inputs #3673 PLC ack #3674 PLC output` • HMI action cmd.reset — applied `#3722 HMI action` • HMI action cmd.auto.off — applied `#3914 HMI action`
Expected behaviour	*No explicit acceptance criterion is recorded in the campaign for this*

experiment.

Observed behaviour

- Wait for Motor_RunCmd = true (up to 5000 ms) — observed `#2523 Observation #2518 PLC output`
- Wait for StackLight_Red = true (up to 8000 ms) — observed `#2653 Observation #2649 PLC output`
- Capture HMI frame — applied `#2783 HMI frame`
- Wait for controller.FaultCode = 0 (up to 8000 ms) — observed `#3733 Observation #3731 PLC output`
- Wait for Motor_RunCmd = false (up to 5000 ms) — observed `#3745 Observation #3743 PLC output`
- Capture HMI frame — applied `#4018 HMI frame`

Cited PLC output `#2745 PLC output` recorded 2026-11-03 23:22:33.156 UTC (source 2026-11-03 23:22:33.128 UTC) · PLC cycle 35355

Motor_RunCmd	false	StackLight_Red	true
controller.ModeText	Faulted	controller.FaultCode	10
controller.FaultTex t	Drive overload	controller.ItemActive	false
controller.CompletionPending	false		

Cited PLC output `#3980 PLC output` recorded 2026-11-03 23:22:52.774 UTC (source 2026-11-03 23:22:52.752 UTC) · PLC cycle 36318

Motor_RunCmd	false	StackLight_Red	false
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ItemActive	false
controller.CompletionPending	false		

Cited PLC output `#4015 PLC output` recorded 2026-11-03 23:22:53.252 UTC (source 2026-11-03 23:22:53.224 UTC) · PLC cycle 36341

Motor_RunCmd	false	StackLight_Red	false
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ItemActive	false
controller.CompletionPending	false		

Recorded conclusion: Observed — conclusion recorded, not a pass verdict

Requirement verdict: UI-01: pass (Muse assessment, unreviewed); UI-02: pass (Muse assessment, unreviewed); UI-03: pass (Muse assessment, unreviewed)

Supporting evidence: `#2508 HMI action #2653 Observation #2745 PLC output #2783 HMI frame #2798 HMI frame #3722 HMI action #3733 Observation #3980 PLC output #4015 PLC output #4075 HMI frame`

Limitations

- FIND-HMI-01: UI-04 stale/unknown/WCS-disconnect behavior was not induced and remains inconclusive; divert-motion feedback congruence was not exercised with a carton.

▾ **EXP-HMI-03** normal Inconclusive

UI-04: Assess stale/unknown/WCS indication only from actually observed healthy HMI/PLC evidence; inconclusive if no stale/disconnect was tested.

Objective	MUSE UI-04: Assess stale/unknown/WCS indication only from actually observed healthy HMI/PLC evidence; inconclusive if no stale/disconnect was tested.
Requirement refs	UI-04
Origin	Planned
Run	2026-11-03 23:23:27.265 UTC → 2026-11-03 23:23:42.705 UTC
Stimulus / operator action	No stimulus or operator action recorded.
Expected behaviour	*No explicit acceptance criterion is recorded in the campaign for this experiment.*
Observed behaviour	No condition checks recorded.

Cited PLC output `#4015 PLC output` recorded 2026-11-03 23:22:53.252 UTC (source 2026-11-03 23:22:53.224 UTC) · PLC cycle 36341

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	1	controller.StartAccepted	true
controller.ResetAccepted	true	controller.EncoderDelta	0

Recorded conclusion	Inconclusive
Requirement verdict	UI-04: inconclusive (Muse assessment, unreviewed)
Supporting evidence	`#5931 Assessment #4075 HMI frame #2798 HMI frame #4015 PLC output #6041 HMI frame`
Limitations	• UI-04 assessed inconclusive: No stale/disconnect stimulus was performed in this UI checkpoint; a healthy Fresh indication cannot prove stale or disconnect handling.

Requirement verdicts

Only verdicts the campaign explicitly recorded; none are inferred from experiment completion

Requirement	Exercised by	Recorded verdict	Separate review artifacts
UI-01	EXP-HMI-01 (observed), EXP-HMI-02 (observed)	pass — Muse assessment, unreviewed ASSESS-UI01-01 (#5922): pass	engineering-review.json (pass_with_limitations, separate artifact): Supported: visible stable controls, applied Start/Stop/Reset and explicit Auto readback limitation.
UI-02	EXP-HMI-01 (observed), EXP-HMI-02 (observed)	pass — Muse assessment, unreviewed ASSESS-UI02-01 (#5925): pass	engineering-review.json (pass_with_limitations, separate artifact): Supported for the captured transitions: RUN command/STOPPED feedback after Start, OFF command/RUNNING feedback during coast after Stop.
UI-03	EXP-HMI-02 (observed)	pass — Muse assessment, unreviewed ASSESS-UI03-01 (#5928): pass	engineering-review.json (pass_with_limitations, separate artifact): Supported by received frame :2798 (3-current.jpg): Faulted, F10 Drive overload, red fault banner and motor off. Extra :2783 was captured but its pixels were not sent to Muse; that citation is excluded from this engineering conclusion.
UI-04	EXP-HMI-03 (inconclusive)	inconclusive — Muse assessment, unreviewed ASSESS-UI04-01 (#5931): inconclusive	engineering-review.json (pass_with_limitations, separate artifact): Inconclusive: no stale/unknown/WCS-disconnect stimulus. No coverage pass claimed.

Findings

Recorded by Muse; observed values are exact PLC outputs, inference is unverified

HMI Start/Stop/Reset and drive-overload fault display matched independent PLC outputs

FIND-HMI-01

Unreviewed

Experiment	EXP-HMI-02
Priority	*Not supplied in the campaign record*
Observed facts	10 exact PLC claims (✓ = matches the cited ledger record)

At `#2745 PLC output` recorded 2026-11-03 23:22:33.156 UTC (source 2026-11-03 23:22:33.128 UTC) · PLC cycle 35355

controller.ModeText	Faulted ✓	controller.FaultText	Drive overload ✓
controller.FaultCode	10 ✓	Motor_RunCmd	false ✓
StackLight_Red	true ✓		

At #4015 PLC output recorded 2026-11-03 23:22:53.252 UTC (source 2026-11-03 23:22:53.224 UTC) · PLC cycle 36341

controller.ModeText	Stopped ✓	controller.FaultCode	0 ✓
Motor_RunCmd	false ✓	controller.ItemActive	false ✓
controller.CompletionPending	false ✓		

Injected fault (truth record) — Injected fault.setOverload {"active":true} #2612 Injected fault

Inference MUSE · UNVERIFIED — After the declared overload injection, PLC independently reported Faulted/Drive overload with motor command off and red stack light on, and HMI pixels showed the same fault/mode on Operation. After clearing the overload and HMI Reset, PLC reported Stopped, fault 0, motor off, no active item and no pending completion, consistent with Reset not restarting the belt.

Assumptions
- Injected truth ref 2612 is kept separate from PLC-output claims.
- No carton sorting was in scope for this UI checkpoint.

Uncertainty — UI-04 stale/unknown/WCS-disconnect behavior was not induced and remains inconclusive; divert-motion feedback congruence was not exercised with a carton.

Engineer review — Unreviewed

HMI behaviour

6 HMI actions recorded; 19 frames captured, 17 of them supplied to Muse; 4 assessments recorded.

Screenshot capture and supply to Muse are separate: each frame states whether its pixels were actually sent to Muse. Assessment verdicts are Muse-authored; any separate review is listed with its own status.

UI-01

Muse verdict: pass · Unreviewed Muse assessment (model-authored) · ASSESS-UI01-01 #5922

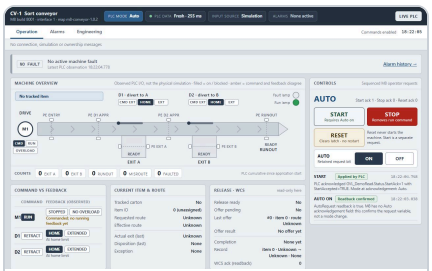
Visible controls / results MUSE — Pixels show stable START, STOP, RESET and AUTO ON/OFF labels. Start frame shows AUTO and 'Applied by PLC'; Stop frame shows STOPPING and 'Applied by PLC'; Reset frame shows STOPPED with Reset 'Applied by PLC' and ResetAck 1 in control header; Auto frames show distinct ON/OFF selected states and 'Readback confirmed'. Independent PLC outputs show StartAck/StopAck then ResetAck=1.

Uncertainty — Auto has readback only, no PLC Auto acknowledgement field as stated on HMI; that limitation was visible and not treated as a mode acknowledgement.

Separate review — engineering-review.json (pass_with_limitations, separate artifact): Supported: visible stable controls, applied Start/Stop/Reset and explicit Auto readback

limitation.

Cited screenshots

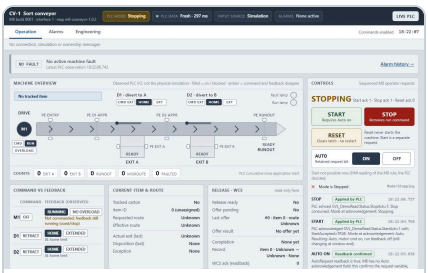


`#1257 HMI frame` captured 2026-11-03 23:22:05.143 UTC

Pixels sent to Muse in request-2 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:20 · sha256 f214117b38606884…`

Embedded from vision-frames/2-2-action.jpg; SHA-256 verified

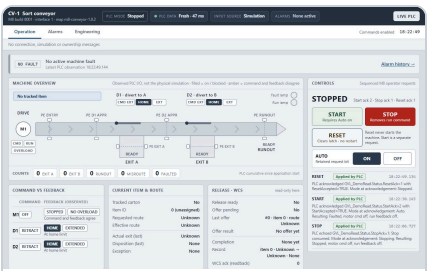


`#1258 HMI frame` captured 2026-11-03 23:22:07.125 UTC

Pixels sent to Muse in request-2 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:23 · sha256 5802d5e287706612…`

Embedded from vision-frames/2-3-action.jpg; SHA-256 verified

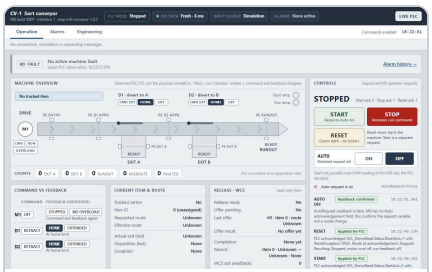


`#4077 HMI frame` captured 2026-11-03 23:22:49.594 UTC

Pixels sent to Muse in request-4 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:43 · sha256 9839fdc5485c5775…`

Embedded from vision-frames/4-2-action.jpg; SHA-256 verified



`#4078 HMI frame` captured 2026-11-03 23:22:51.957 UTC

Pixels sent to Muse in request-4 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:45 · sha256 38094356559f9b74…`

Embedded from vision-frames/4-3-action.jpg; SHA-256 verified

Cited actions and acknowledgements

HMI action	Input delivered	PLC acknowledgement	Rendered result
`#798 cmd.auto.on` Auto request on	2026-11-03 23:22:02.991 UTC Trusted page input	applied (readback) — AutoRequest readback is true. M8 has no Auto acknowledgement field: this confirms the request variable, not a mode change. 2026-11-03 23:22:03.038 UTC	page rendered: Auto on · Readback confirmed at 2026-11-03T23:22:03.087Z screen `#1256 HMI frame` sent to Muse

HMI action	Input delivered	PLC acknowledgement	Rendered result
`#938 cmd.start` START: Requires Auto on	2026-11-03 23:22:04.668 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.StartAck=1 with StartAccepted=TRUE. Mode at acknowledgement: Auto. 2026-11-03 23:22:04.748 UTC	page rendered: Start · Applied by PLC at 2026-11-03T23:22:04.804Z screen `#1257 HMI frame` sent to Muse
`#1120 cmd.stop` STOP: Removes run command	2026-11-03 23:22:06.627 UTC Trusted page input	applied (plc sequence echo) — PLC echoed GVL_DemoRead.Status.StopAck=1: Stop consumed. Mode at acknowledgement: Stopping. 2026-11-03 23:22:06.727 UTC	page rendered: Stop · Applied by PLC at 2026-11-03T23:22:06.790Z screen `#1258 HMI frame` sent to Muse
`#3722 cmd.reset` RESET: Clears latch · no restart	2026-11-03 23:22:49.047 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.ResetAck=1 with ResetAccepted=TRUE. Mode at acknowledgement: Stopped. 2026-11-03 23:22:49.134 UTC	page rendered: Reset · Applied by PLC at 2026-11-03T23:22:49.201Z screen `#4077 HMI frame` sent to Muse
`#3914 cmd.auto.off` Auto request off	2026-11-03 23:22:51.458 UTC Trusted page input	applied (readback) — AutoRequest readback is false. M8 has no Auto acknowledgement field: this confirms the request variable, not a mode change. 2026-11-03 23:22:51.541 UTC	page rendered: Auto off · Readback confirmed at 2026-11-03T23:22:51.573Z screen `#4078 HMI frame` sent to Muse

Matching PLC observations

▼ `#1197 PLC output` recorded 2026-11-03 23:22:08.003 UTC · PLC cycle 34122

At `#1197 PLC output` recorded 2026-11-03 23:22:08.003 UTC (source 2026-11-03 23:22:07.979 UTC) · PLC cycle 34122

`Motor_RunCmd`	`false`	`D1_ExtendCmd`	`false`
`D2_ExtendCmd`	`false`	`StackLight_Red`	`false`
`StackLight_Green`	`false`	`controller.Mode`	`0`
`controller.ModeText`	`Stopped`	`controller.FaultCode`	`0`
`controller.FaultText`	`""`	`controller.ExceptionCode`	`0`
`controller.ExceptionText`	`""`	`controller.ReleaseReady`	`false`
`controller.ItemActive`	`false`	`controller.ActiveItemId`	`0`
`controller.RequestedRoute`	`0`	`controller.EffectiveRoute`	`0`
`controller.ActualExit`	`0`	`controller.Disposition`	`0`
`controller.OfferAckSequence`	`0`	`controller.OfferAckItemId`	`0`

controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	1	controller.StopAck	1
controller.ResetAck	0	controller.StartAccepted	true
controller.ResetAccepted	false	controller.EncoderDelta	0

▼ #3980 PLC output recorded 2026-11-03 23:22:52.774 UTC · PLC cycle 36318

At #3980 PLC output recorded 2026-11-03 23:22:52.774 UTC (source 2026-11-03 23:22:52.752 UTC) · PLC cycle 36318

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	1	controller.StartAccepted	true
controller.ResetAccepted	true	controller.EncoderDelta	0

UI-02 Muse verdict: pass · Unreviewed Muse assessment (model-authored) · ASSESS-UI02-01 #5925

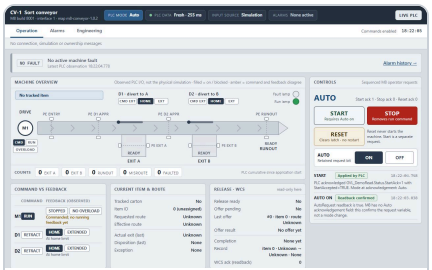
Visible controls / results

MUSE

Operation view has separate COMMAND and FEEDBACK (OBSERVED) columns. Start-result pixels show M1 RUN commanded while feedback is STOPPED with text 'Commanded; no running feedback yet'. Stop-result pixels show M1 OFF while feedback is RUNNING with text 'Not commanded; feedback still running (coast/stop)'. Diverts show RETRACT command

	separately from HOME/EXTENDED feedback. This distinguishes command from feedback rather than painting running solely from command.
Uncertainty	Assessment is for the displayed command-versus-feedback distinction in the captured Start/Stop transitions; no carton/divert motion sequence was in scope.
Separate review	engineering-review.json (pass_with_limitations, separate artifact): Supported for the captured transitions: RUN command/STOPPED feedback after Start, OFF command/RUNNING feedback during coast after Stop.

Cited screenshots

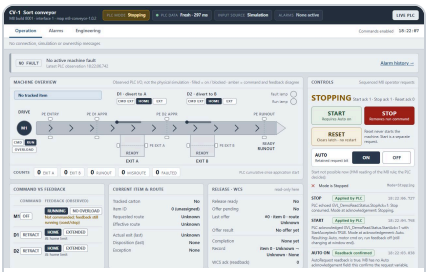


`#1257 HMI frame` captured 2026-11-03 23:22:05.143 UTC

Pixels sent to Muse in request-2 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:20 · sha256 f214117b38606884…`

Embedded from vision-frames/2-2-action.jpg; SHA-256 verified

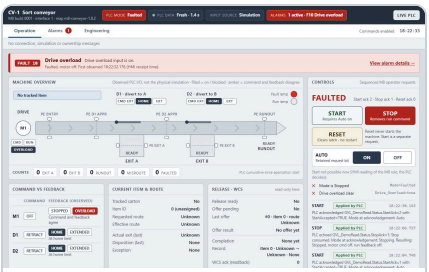


`#1258 HMI frame` captured 2026-11-03 23:22:07.125 UTC

Pixels sent to Muse in request-2 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:23 · sha256 5802d5e287706612…`

Embedded from vision-frames/2-3-action.jpg; SHA-256 verified



`#2798 HMI frame` captured 2026-11-03 23:22:33.791 UTC

Pixels sent to Muse in request-3 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:36 · sha256 216f9f7ddb242a03…`

Embedded from vision-frames/3-current.jpg; SHA-256 verified

Cited actions and acknowledgements

HMI action	Input delivered	PLC acknowledgement	Rendered result
`#938 cmd.start` START: Requires Auto on	2026-11-03 23:22:04.668 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.StartAck=1 with StartAccepted=TRUE. Mode at acknowledgement: Auto. 2026-11-03 23:22:04.748 UTC	page rendered: Start · Applied by PLC at 2026-11-03T23:22:04.804Z screen `#1257 HMI frame` sent to Muse
`#1120 cmd.stop` STOP: Removes run command	2026-11-03 23:22:06.627 UTC Trusted page input	applied (plc sequence echo) — PLC echoed GVL_DemoRead.Status.StopAck=1: Stop consumed. Mode at acknowledgement: Stopping. 2026-11-03 23:22:06.727 UTC	page rendered: Stop · Applied by PLC at 2026-11-03T23:22:06.790Z screen `#1258 HMI frame` sent to Muse

Matching PLC observations

▼ `#2745 PLC output` recorded 2026-11-03 23:22:33.156 UTC · PLC cycle 35355

At #2745 PLC output recorded 2026-11-03 23:22:33.156 UTC (source 2026-11-03 23:22:33.128 UTC) · PLC cycle 35355

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	true
StackLight_Green	false	controller.Mode	3
controller.ModeText	Faulted	controller.FaultCode	10
controller.FaultText	Drive overload	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	0	controller.StartAccepted	true
controller.ResetAccepted	false	controller.EncoderDelta	0

▼ #1197 PLC output recorded 2026-11-03 23:22:08.003 UTC · PLC cycle 34122

At #1197 PLC output recorded 2026-11-03 23:22:08.003 UTC (source 2026-11-03 23:22:07.979 UTC) · PLC cycle 34122

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	1	controller.StopAck	1

controller.ResetAck	0	controller.StartAccepted	true
controller.ResetAccepted	false	controller.EncoderDelta	0

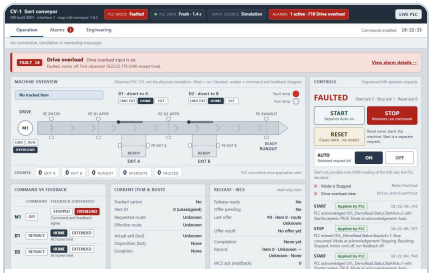
UI-03 Muse verdict: pass | Unreviewed Muse assessment (model-authored) | ASSESS-UI03-01 #5928

Visible controls / results MUSE	On the main Operation view without navigation, pixels show top banner 'PLC MODE Faulted', 'ALARMS 1 active - F10 Drive overload', red banner 'FAULT 10 Drive overload - Drive overload input is on. Faulted, motor off', Alarms tab badge 1, and CONTROLS heading FAULTED. Independent PLC output at 2745 is ModeText Faulted, FaultCode 10, FaultText Drive overload.
Uncertainty	Only the injected drive-overload fault text was tested; other fault texts were not exercised.
Separate review	engineering-review.json (pass_with_limitations, separate artifact): Supported by received frame :2798 (3-current.jpg): Faulted, F10 Drive overload, red fault banner and motor off. Extra :2783 was captured but its pixels were not sent to Muse; that citation is excluded from this engineering conclusion.

Citation check

- Cites #2783: pixels were not among the images sent to Muse (model exchange log); also flagged by s2-audit.json and engineering-review.json.

Cited screenshots



`#2798 HMI frame` captured 2026-11-03 23:22:33.791 UTC

Pixels sent to Muse in request-3 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:36 · sha256 216f9f7ddb242a03…`

Embedded from vision-frames/3-current.jpg; SHA-256 verified

Image unavailable

Captured in the HMI session; image bytes are not preserved in this evidence package

`#2783 · sha256 542d79c5c6506de0…`

`#2783 HMI frame` captured 2026-11-03 23:22:33.613 UTC

Captured only: pixels were not among the images sent to Muse (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:35 · sha256 542d79c5c6506de0…`

Cited actions and acknowledgements

HMI action	Input delivered	PLC acknowledgement	Rendered result
`#2508 cmd.start` START: Requires Auto on	2026-11-03 23:22:29.954 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.StartAck=2 with StartAccepted=TRUE. Mode at acknowledgement: Auto. 2026-11-03 23:22:30.143 UTC	page rendered: Start · Applied by PLC at 2026-11-03T23:22:30.216Z screen `#2801 HMI frame` sent to Muse

Matching PLC observations

▼ `#2745 PLC output` recorded 2026-11-03 23:22:33.156 UTC · PLC cycle 35355

At `#2745 PLC output` recorded 2026-11-03 23:22:33.156 UTC (source 2026-11-03 23:22:33.128 UTC) · PLC cycle 35355

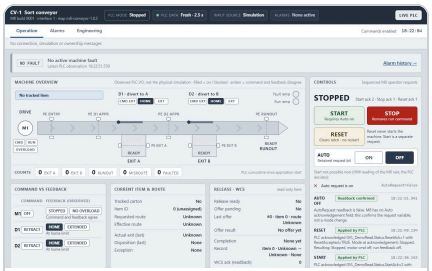
Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	true
StackLight_Green	false	controller.Mode	3
controller.ModeText	Faulted	controller.FaultCode	10
controller.FaultText	Drive overload	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	0	controller.StartAccepted	true
controller.ResetAccepted	false	controller.EncoderDelta	0

UI-04 Muse verdict: inconclusive · Unreviewed Muse assessment (model-authored) · `ASSESS-UI04-01` #5931

Visible controls / results MUSE	Healthy/fault frames show 'PLC DATA Fresh' with an age and a RELEASE-WCS panel, but no stale-data condition, unknown-quality condition, or WCS disconnect was induced or observed. Therefore stale/unknown and WCS-disconnect display behavior was not tested.
Uncertainty	No stale/disconnect stimulus was performed in this UI checkpoint; a healthy Fresh indication cannot prove stale or disconnect handling.

Separate review	engineering-review.json (pass_with_limitations, separate artifact): Inconclusive: no stale/unknown/WCS-disconnect stimulus. No coverage pass claimed.

Cited screenshots

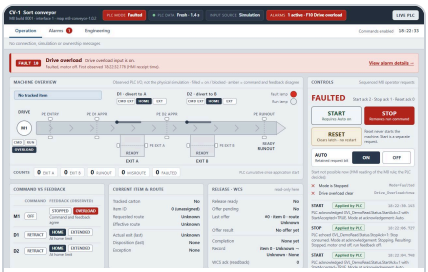


`#4075 HMI frame` captured 2026-11-03 23:22:54.290 UTC

Pixels sent to Muse in request-4 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:47 · sha256 2dde6e823dfc3977…`

Embedded from vision-frames/4-current.jpg; SHA-256 verified



`#2798 HMI frame` captured 2026-11-03 23:22:33.791 UTC

Pixels sent to Muse in request-3 (model exchange log)

`hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:36 · sha256 216f9f7ddb242a03…`

Embedded from vision-frames/3-current.jpg; SHA-256 verified

Matching PLC observations

▼ `#4015 PLC output` recorded 2026-11-03 23:22:53.252 UTC · PLC cycle 36341

At `#4015 PLC output` recorded 2026-11-03 23:22:53.252 UTC (source 2026-11-03 23:22:53.224 UTC) · PLC cycle 36341

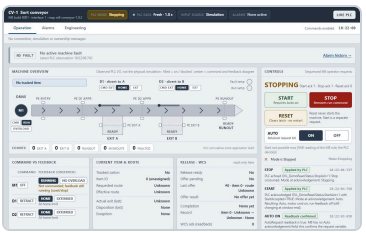
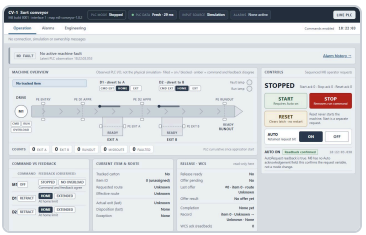
Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	1	controller.StartAccepted	true
controller.ResetAccepted	true	controller.EncoderDelta	0

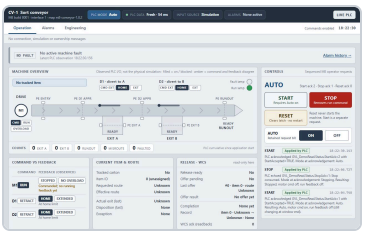
▼ All HMI actions (6)

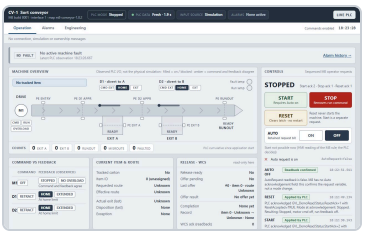
HMI action	Input delivered	PLC acknowledgement	Rendered result
#798 cmd.auto.on Auto request on	2026-11-03 23:22:02.991 UTC Trusted page input	applied (readback) — AutoRequest readback is true. M8 has no Auto acknowledgement field: this confirms the request variable, not a mode change. 2026-11-03 23:22:03.038 UTC	page rendered: Auto on · Readback confirmed at 2026-11-03T23:22:03.087Z screen `#1256 HMI frame` sent to Muse
#938 cmd.start START: Requires Auto on	2026-11-03 23:22:04.668 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.StartAck=1 with StartAccepted=TRUE. Mode at acknowledgement: Auto. 2026-11-03 23:22:04.748 UTC	page rendered: Start · Applied by PLC at 2026-11-03T23:22:04.804Z screen `#1257 HMI frame` sent to Muse
#1120 cmd.stop STOP: Removes run command	2026-11-03 23:22:06.627 UTC Trusted page input	applied (plc sequence echo) — PLC echoed GVL_DemoRead.Status.StopAck=1: Stop consumed. Mode at acknowledgement: Stopping. 2026-11-03 23:22:06.727 UTC	page rendered: Stop · Applied by PLC at 2026-11-03T23:22:06.790Z screen `#1258 HMI frame` sent to Muse

HMI action	Input delivered	PLC acknowledgement	Rendered result
`#2508 cmd.start` START: Requires Auto on	2026-11-03 23:22:29.954 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.StartAck=2 with StartAccepted=TRUE. Mode at acknowledgement: Auto. 2026-11-03 23:22:30.143 UTC	page rendered: Start · Applied by PLC at 2026-11-03T23:22:30.216Z screen `#2801 HMI frame` sent to Muse
`#3722 cmd.reset` RESET: Clears latch · no restart	2026-11-03 23:22:49.047 UTC Trusted page input	applied (plc sequence ack) — PLC acknowledged GVL_DemoRead.Status.ResetAck=1 with ResetAccepted=TRUE. Mode at acknowledgement: Stopped. 2026-11-03 23:22:49.134 UTC	page rendered: Reset · Applied by PLC at 2026-11-03T23:22:49.201Z screen `#4077 HMI frame` sent to Muse
`#3914 cmd.auto.off` Auto request off	2026-11-03 23:22:51.458 UTC Trusted page input	applied (readback) — AutoRequest readback is false. M8 has no Auto acknowledgement field: this confirms the request variable, not a mode change. 2026-11-03 23:22:51.541 UTC	page rendered: Auto off · Readback confirmed at 2026-11-03T23:22:51.573Z screen `#4078 HMI frame` sent to Muse

▼ All captured frames (19; 17 supplied to Muse)

Frame	Captured	Supplied to Muse	Image
`#43 vision-1-current`	2026-11-03 23:21:50.181 UTC	Pixels sent to Muse in request-1 (model exchange log)	Not selected for this report; metadata only
`#1255 vision-2-current`	2026-11-03 23:22:08.727 UTC	Pixels sent to Muse in request-2 (model exchange log)	
`#1256 vision-2-action`	2026-11-03 23:22:03.416 UTC	Pixels sent to Muse in request-2 (model exchange log)	
`#1257 vision-2-action`	2026-11-03 23:22:05.143 UTC	Pixels sent to Muse in request-2 (model exchange log)	shown with its assessment above

Frame	Captured	Supplied to Muse	Image
`#1258 vision-2-action`	2026-11-03 23:22:07.125 UTC	Pixels sent to Muse in request-2 (model exchange log)	shown with its assessment above
`#2783 cap-fault-01`	2026-11-03 23:22:33.613 UTC	Captured only: pixels were not among the images sent to Muse (model exchange log)	Captured in the HMI session; image bytes are not preserved in this evidence package
`#2798 vision-3-current`	2026-11-03 23:22:33.791 UTC	Pixels sent to Muse in request-3 (model exchange log)	shown with its assessment above
`#2799 vision-3-action`	2026-11-03 23:22:05.143 UTC	Pixels sent to Muse in request-3 (model exchange log)	Not selected for this report; metadata only
`#2800 vision-3-action`	2026-11-03 23:22:07.125 UTC	Pixels sent to Muse in request-3 (model exchange log)	Not selected for this report; metadata only
`#2801 vision-3-action`	2026-11-03 23:22:30.563 UTC	Pixels sent to Muse in request-3 (model exchange log)	
`#4018 cap-final-01`	2026-11-03 23:22:53.290 UTC	Captured only: pixels were not among the images sent to Muse (model exchange log)	Not selected for this report; metadata only
`#4075 vision-4-current`	2026-11-03 23:22:54.290 UTC	Pixels sent to Muse in request-4 (model exchange log)	shown with its assessment above
`#4076 vision-4-action`	2026-11-03 23:22:30.563 UTC	Pixels sent to Muse in request-4 (model exchange log)	Not selected for this report; metadata only
`#4077 vision-4-action`	2026-11-03 23:22:49.594 UTC	Pixels sent to Muse in request-4 (model exchange log)	shown with its assessment above
`#4078 vision-4-action`	2026-11-03 23:22:51.957 UTC	Pixels sent to Muse in request-4 (model exchange log)	shown with its assessment above

Frame	Captured	Supplied to Muse	Image
`#6041 vision-5-current`	2026-11-03 23:23:28.820 UTC	Pixels sent to Muse in request-5 (model exchange log)	
`#6042 vision-5-action`	2026-11-03 23:22:30.563 UTC	Pixels sent to Muse in request-5 (model exchange log)	Not selected for this report; metadata only
`#6043 vision-5-action`	2026-11-03 23:22:49.594 UTC	Pixels sent to Muse in request-5 (model exchange log)	Not selected for this report; metadata only
`#6044 vision-5-action`	2026-11-03 23:22:51.957 UTC	Pixels sent to Muse in request-5 (model exchange log)	Not selected for this report; metadata only

Recommended next checks

RECOMMENDATIONS GENERATED FROM GAPS IN THIS RECORD · NOT TEST RESULTS

1. Define and apply a stimulus that exercises UI-04, then re-run EXP-HMI-03; its recorded conclusion is inconclusive.
2. Re-review ASSESS-UI03-01 (UI-03) against only the frames Muse actually received.
3. Engineer review: accept or request a rerun for each unreviewed finding and assessment.
4. Recover or re-capture 1 cited image missing from this package.

Remaining work, missing evidence and limits

Inconclusive

- EXP-HMI-03: UI-04: Assess stale/unknown/WCS indication only from actually observed healthy HMI/PLC evidence; inconclusive if no stale/disconnect was tested.
- UI-04 (ASSESS-UI04-01): No stale/disconnect stimulus was performed in this UI checkpoint; a healthy Fresh indication cannot prove stale or disconnect handling.

Missing evidence

- Image 542d79c5c650…: Captured in the HMI session; image bytes are not preserved in this evidence package

Scope limits

- Wonder-directed M8 campaign through the runtime owner's verified isolated adapter. Scope-limited experiment evidence; no certified controller coverage. HMI frames remain separate.
- Experiment evidence only; no final conveyor-controller requirement coverage is certified.

Final recorded PLC and connection state

historical

Last PLC output recorded in this campaign, at 2026-11-03 23:23:43.767 UTC. Historical: this report does not monitor current PLC state or freshness.

PLC output recorded	2026-11-03 23:23:43.767 UTC (source 2026-11-03 23:23:43.418 UTC)
PLC cycle	38803
PLC boot identity	1787886077538
Connector state	fresh — Observed simulation active under stable PLC boot (recorded 2026-11-03 23:23:43.745 UTC)

▼ All 40 recorded output values

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	0	controller.Disposition	0
controller.OfferAckSequence	0	controller.OfferAckItemId	0
controller.OfferAccepted	false	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	0
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	0
controller.CompletionDisposition	0	controller.CountExitA	0
controller.CountExitB	0	controller.CountRunout	0
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	2	controller.StopAck	1
controller.ResetAck	1	controller.StartAccepted	true
controller.ResetAccepted	true	controller.EncoderDelta	0

Usage and accounting

Reservations and estimates, not invoices

Budget reservations

This campaign	$25.00
Session total (includes earlier campaigns)	$135.00
Held for planning	$5.00
Held for EXP-HMI-01	$5.00
Held for EXP-HMI-02	$10.00
Held for EXP-HMI-03	$5.00

Conservative reservations are admission accounting held before each model request. They are not spend and not an invoice.

Model token usage (as reported per request)

Model requests	5 (5 completed)
Input tokens	355,185
of which cached	159,068
Output tokens	7,653
of which reasoning	3,335

Token-cost estimates

Estimate at uncached list rates (5 of 5 requests priced)	≈ $1.5738
Estimate stated in s2-audit.json	≈ $1.5738
Estimate stated in engineering-review.json	≈ $1.5738

Token-cost estimates use the pricing recorded with each request (4/20 USD per million input/output tokens, checked 2026-11-01). Estimates, not the invoice; caching and billing can differ.

Separate review artifacts

Not part of the model report; status and origin shown as recorded

▾ **`s2-audit.json`** offline audit status: reference review required

Separate offline audit of the saved evidence. Not part of the model report; it does not change any recorded conclusion.

Recorded	2026-11-03 23:29:38.249 UTC

File SHA-256	`3d1e92420fa8aff8c578a7ec6b9144f87c080cc6cd47d77566878331dee7981e`
Original evidence hashes verified	24
Paired tool results	28
Exact PLC finding claims checked	10
Images verified	17
HMI actions verified	4
Expected HMI actions missing	none
Expected assessments missing	none
All assessment frame citations were viewed by Muse	no
Model requests	5
Recorded failures	none
Final recorded PLC state clean	yes
In-flight commands at close	none
Citation defects	portal-20261103232137992-1:5928 cites portal-20261103232137992-1:2783
Claim boundary	Exact frames were sent to Muse; trusted browser inputs, rendered outcomes and PLC acknowledgements are preserved. Pixel interpretations require separate review; inconclusive UI coverage remains inconclusive.

▾ **`engineering-review.json`** engineering review status: pass with limitations

Separate engineering review artifact. Not part of the model report; the original Muse findings and assessments remain unreviewed in the campaign record.

Scope	Muse browser HMI operation and UI-01/UI-02/UI-03 screen checks against live M8
Recorded	2026-11-03 23:30:48.076 UTC
File SHA-256	`6c084c8fd3a40b1c7b9c892319ba766cd4f91e08b47d7acf818e24e8dad1e251`
Claims traced to original child/transport records	10
Model requests	5
All model citations valid	no

Frames reviewed	4
Earlier attempt	The preceding one-request attempt stopped on a real PLC heartbeat lapse before any HMI action. It remains partial. Retry reduced observer load only; physical engine, world, PLC and timing bounds were unchanged.
Final PLC check	Fresh same-boot read confirms Stopped, fault0, motor off, no item/offer/completion; retained simulation image at rest. ConnectorStale after child exit is expected bench diagnostic.
Runtime ownership released	yes
Citation defects	portal-20261103232137992-1:5928 cites portal-20261103232137992-1:2783
UI-01	Supported: visible stable controls, applied Start/Stop/Reset and explicit Auto readback limitation.
UI-02	Supported for the captured transitions: RUN command/STOPPED feedback after Start, OFF command/RUNNING feedback during coast after Stop.
UI-03	Supported by received frame :2798 (3-current.jpg): Faulted, F10 Drive overload, red fault banner and motor off. Extra :2783 was captured but its pixels were not sent to Muse; that citation is excluded from this engineering conclusion.
UI-04	Inconclusive: no stale/unknown/WCS-disconnect stimulus. No coverage pass claimed.
Claim boundary	Scoped HMI operation proof passed with a report citation defect. Not all HMI requirements passed; not arbitrary-HMI navigation, native CODESYS WebVisu, exhaustive alarms or physical hardware validation.

Evidence appendix

77 cited records of 6,889 ledger events; the full ledger is a separate file

Integrity and sources

Report	`report.json sha256 009e91dd1efd7158c25e3f671fee5f951bb584c0f792efc22da9a2651237a06d`
Matches reconstruction from ledger	yes
Ledger	`6,889 events · head c91d59a42b413ccc… · tail 6b539d24f7ed9c63…`
Recorded span	2026-11-03 23:21:38.139 UTC → 2026-11-03 23:23:43.818 UTC
Report generated	2026-11-04 00:37:57.280 UTC

Source file	Role	SHA-256	Size
`report.json`	Original structured campaign	`009e91dd1efd7158c25e3f671fee5f951bb584c`	109.8

Source file	Role	SHA-256	Size
	report (unchanged)	0f792efc22da9a2651237a06d	KiB
campaign.jsonl	Full hash-chained campaign ledger · 6,889 events	2e43736f0c0f10ba82ff4b762b69e944ae572d3b8008b28e4adf407034067ded	12090.6 KiB
s2-audit.json	Separate offline audit	3d1e92420fa8aff8c578a7ec6b9144f87c080cc6cd47d77566878331dee7981e	6.2 KiB
engineering-review.json	Separate engineering review	6c084c8fd3a40b1c7b9c892319ba766cd4f91e08b47d7acf818e24e8dad1e251	5.9 KiB
wonder-exchanges.jsonl	Model request/response and image-input lineage	5d566f3b6b2c900c8a0bf5a10e7e57a3303cf0bf6007f7b6a31766b3c1c37fb5	3696.3 KiB
vision-frames/2-current.jpg	HMI image (verified against ledger hash)	2f75d6515c0782e2f7e58302502f0a8734c39a220e0517ce688801cb4d6c4c78	144.0 KiB
vision-frames/2-1-action.jpg	HMI image (verified against ledger hash)	ad0b8d292d1068b54050b43fc9aac501ad88bf25a8e18258c4789aa77529d211	124.2 KiB
vision-frames/2-2-action.jpg	HMI image (verified against ledger hash)	f214117b3860688481fb65635b72c81c86fc2c6963e90625ad5cd6c162e15844	132.0 KiB
vision-frames/2-3-action.jpg	HMI image (verified against ledger hash)	5802d5e287706612b2b7d701222409c76cf148b7e4f1fa426552d1636923f748	144.1 KiB
vision-frames/3-current.jpg	HMI image (verified against ledger hash)	216f9f7ddb242a03f80bf5789c97c81f5b962672735147aef4958ac7d876004d	147.7 KiB
vision-frames/3-3-action.jpg	HMI image (verified against ledger hash)	b60401e719747c576de53571665c2de1b6feeaab0ab1d173c0526207d6c79c5a	140.7 KiB
vision-frames/4-current.jpg	HMI image (verified against ledger hash)	2dde6e823dfc39779a12a2e662d1e3d313a669e7af05a10586ecaa2d296cbecf	140.2 KiB
vision-frames/4-2-action.jpg	HMI image (verified against ledger hash)	9839fdc5485c577507a4137d34114d779c99a1d4443f5119b25f237fe3160cee	139.3 KiB
vision-frames/4-3-action.jpg	HMI image (verified against ledger hash)	38094356559f9b74a35d72ed82fec564f8e8439487f3e8e227a145ee42ec92ef	140.2 KiB
vision-frames/5-current.jpg	HMI image (verified against ledger hash)	496b6229ab2914559fdc64d8660730a33e821fb84116da4c38ac81b874ded1c7	140.2 KiB

▼ Muse decision trail (5 model requests; decisions are model-authored)

Request	Started · elapsed	Tokens	Decision summary (Muse)
request-1 completed planning	2026-11-03 23:21:50.075 UTC 13 s	16,879 in (0 cached) · 1,443 out response returned · 1 image input	Plan UI checkpoint and execute first HMI sequence: Auto On, Start, observe PLC motor command, then Stop. No cartons.
request-2 completed EXP-HMI-01	2026-11-03 23:22:08.619 UTC 21 s	44,598 in (16,756 cached) · 1,338 out response returned · 4 image inputs	Complete EXP-HMI-01 from Start/Stop evidence, then start again via HMI and inject drive overload to test fault display and Reset.
request-3 completed EXP-HMI-02	2026-11-03 23:22:33.657 UTC 15 s	70,607 in (44,532 cached) · 965 out response returned · 4 image inputs	Clear overload fault, perform HMI Reset, verify fault clears with motor off, then turn Auto off and capture final frame.
request-4 completed EXP-HMI-02	2026-11-03 23:22:53.320 UTC 34 s	97,810 in (0 cached) · 3,306 out response returned · 4 image inputs	Complete EXP-HMI-02 and record UI-01..UI-04 assessments plus grounded finding. EXP-HMI-03 remains to complete next from healthy-frame evidence with UI-04 inconclusive.
request-5 completed EXP-HMI-03	2026-11-03 23:23:28.021 UTC 15 s	125,291 in (97,780 cached) · 601 out response returned · 4 image inputs	Complete UI-04 experiment as inconclusive (no stale/disconnect induced) and finish campaign Stopped fault-free.

Cited evidence records

▼ Evidence #43 – #3731 (50 records)

Evidence	Recorded	Summary
#43 HMI frame	2026-11-03 23:21:50.194 UTC source 2026-11-03 23:21:50.181 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:12 `portal-20261103232137992-1:43`
#798 HMI action	2026-11-03 23:22:03.425 UTC source 2026-11-03 23:22:03.416 UTC	HMI action cmd.auto.on · PLC applied `portal-20261103232137992-1:798`

#807
PLC output
	2026-11-03 23:22:03.478 UTC
	source 2026-11-03 23:22:03.431 UTC
		PLC outputs at cycle 33899
		`cycle 33899 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:807`

#895
PLC output
	2026-11-03 23:22:04.642 UTC
	source 2026-11-03 23:22:04.618 UTC
		PLC outputs at cycle 33957
		`cycle 33957 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:895`

#897 Observation	2026-11-03 23:22:04.653 UTC	Observation window 1200 ms completed `portal-20261103232137992-1:897`
#938 HMI action	2026-11-03 23:22:05.150 UTC source 2026-11-03 23:22:05.143 UTC	HMI action cmd.start · PLC applied `portal-20261103232137992-1:938`

#947
PLC output
	2026-11-03 23:22:05.362 UTC
	source 2026-11-03 23:22:05.167 UTC
		PLC outputs at cycle 33984
		`cycle 33984 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:947`

#952 Observation	2026-11-03 23:22:05.386 UTC	Observed Motor_RunCmd = true `portal-20261103232137992-1:952`

#962
PLC output
	2026-11-03 23:22:05.434 UTC
	source 2026-11-03 23:22:05.269 UTC
		PLC outputs at cycle 33989
		`cycle 33989 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:962`

#1072
PLC output
	2026-11-03 23:22:06.602 UTC
	source 2026-11-03 23:22:06.578 UTC
		PLC outputs at cycle 34053
		`cycle 34053 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:1072`

Evidence	Recorded	Summary
#1073 Observation	2026-11-03 23:22:06.608 UTC	Observation window 1200 ms completed `portal-20261103232137992-1:1073`
#1120 HMI action	2026-11-03 23:22:07.134 UTC source 2026-11-03 23:22:07.125 UTC	HMI action cmd.stop · PLC applied `portal-20261103232137992-1:1120`

#1129
PLC output
	2026-11-03 23:22:07.176 UTC
	source 2026-11-03 23:22:07.127 UTC
		PLC outputs at cycle 34080
		`cycle 34080 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:1129`

#1131 Observation	2026-11-03 23:22:07.380 UTC	Observed Motor_RunCmd = false `portal-20261103232137992-1:1131`

#1141
PLC output
	2026-11-03 23:22:07.430 UTC
	source 2026-11-03 23:22:07.229 UTC
		PLC outputs at cycle 34085
		`cycle 34085 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:1141`

#1197
PLC output
	2026-11-03 23:22:08.003 UTC
	source 2026-11-03 23:22:07.979 UTC
		PLC outputs at cycle 34122
		`cycle 34122 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:1197`

#1238
PLC output
	2026-11-03 23:22:08.584 UTC
	source 2026-11-03 23:22:08.556 UTC
		PLC outputs at cycle 34150
		`cycle 34150 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:1238`

#1240 Observation	2026-11-03 23:22:08.599 UTC	Observation window 1200 ms completed `portal-20261103232137992-1:1240`
#1255 HMI frame	2026-11-03 23:22:08.739 UTC source 2026-11-03 23:22:08.727 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:24 `portal-20261103232137992-1:1255`
#1256 HMI frame	2026-11-03 23:22:08.776 UTC source 2026-11-03 23:22:03.416 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:17 `portal-20261103232137992-1:1256`
#1257 HMI frame	2026-11-03 23:22:08.785 UTC source 2026-11-03 23:22:05.143 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:20 `portal-20261103232137992-1:1257`
#1258 HMI frame	2026-11-03 23:22:08.796 UTC source 2026-11-03 23:22:07.125 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:23 `portal-20261103232137992-1:1258`

Evidence	Recorded	Summary
#2508 HMI action	2026-11-03 23:22:30.574 UTC source 2026-11-03 23:22:30.563 UTC	HMI action cmd.start · PLC applied `portal-20261103232137992-1:2508`

#2518
PLC output
	2026-11-03 23:22:30.632 UTC
	source 2026-11-03 23:22:30.608 UTC
		PLC outputs at cycle 35231
		`cycle 35231 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2518`

#2523 Observation	2026-11-03 23:22:30.658 UTC	Observed Motor_RunCmd = true `portal-20261103232137992-1:2523`

#2533
PLC output
	2026-11-03 23:22:30.758 UTC
	source 2026-11-03 23:22:30.730 UTC
		PLC outputs at cycle 35237
		`cycle 35237 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2533`

#2605
PLC output
	2026-11-03 23:22:31.655 UTC
	source 2026-11-03 23:22:31.625 UTC
		PLC outputs at cycle 35281
		`cycle 35281 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2605`

#2608 Observation	2026-11-03 23:22:31.682 UTC	Observation window 1000 ms completed `portal-20261103232137992-1:2608`
#2612 Injected fault	2026-11-03 23:22:31.707 UTC	Injected fault.setOverload {"active":true} `portal-20261103232137992-1:2612`
#2627 Source record	2026-11-03 23:22:31.840 UTC	Coordinator connector commit.acknowledged `portal-20261103232137992-1:2627`

#2628
Raw inputs
	2026-11-03 23:22:31.846 UTC
	source 2026-11-03 23:22:31.822 UTC
		Raw input image, commit sequence 384
		`cycle 35291 · boot 1787886077538`
		▶ 20 values
		`portal-20261103232137992-1:2628`

#2629 PLC ack	2026-11-03 23:22:31.850 UTC	PLC applied commit 384 (52.9 ms) `portal-20261103232137992-1:2629`

#2630
PLC output
	2026-11-03 23:22:31.854 UTC
	source 2026-11-03 23:22:31.822 UTC
		PLC outputs at cycle 35291
		`cycle 35291 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2630`

Evidence	Recorded	Summary

#2649
PLC output
	2026-11-03 23:22:31.962 UTC
	source 2026-11-03 23:22:31.927 UTC
		PLC outputs at cycle 35296
		`cycle 35296 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2649`

#2653 Observation	2026-11-03 23:22:31.986 UTC	Observed StackLight_Red = true `portal-20261103232137992-1:2653`

#2663
PLC output
	2026-11-03 23:22:32.099 UTC
	source 2026-11-03 23:22:32.067 UTC
		PLC outputs at cycle 35303
		`cycle 35303 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2663`

#2745
PLC output
	2026-11-03 23:22:33.156 UTC
	source 2026-11-03 23:22:33.128 UTC
		PLC outputs at cycle 35355
		`cycle 35355 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2745`

#2761
PLC output
	2026-11-03 23:22:33.478 UTC
	source 2026-11-03 23:22:33.333 UTC
		PLC outputs at cycle 35365
		`cycle 35365 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:2761`

#2766 Observation	2026-11-03 23:22:33.509 UTC	Observation window 1500 ms completed `portal-20261103232137992-1:2766`
#2783 HMI frame	2026-11-03 23:22:33.632 UTC source 2026-11-03 23:22:33.613 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:35 `portal-20261103232137992-1:2783`
#2798 HMI frame	2026-11-03 23:22:33.804 UTC source 2026-11-03 23:22:33.791 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:36 `portal-20261103232137992-1:2798`
#2799 HMI frame	2026-11-03 23:22:33.905 UTC source 2026-11-03 23:22:05.143 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:20 `portal-20261103232137992-1:2799`
#2800 HMI frame	2026-11-03 23:22:33.917 UTC source 2026-11-03 23:22:07.125 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:23 `portal-20261103232137992-1:2800`
#2801 HMI frame	2026-11-03 23:22:33.925 UTC source 2026-11-03 23:22:30.563 UTC	HMI frame hmi-frame:hmi-e926e4dd-50e5-4844-936d-6b8d8b7d48f4:33 `portal-20261103232137992-1:2801`

Evidence	Recorded	Summary
#3671 Source record	2026-11-03 23:22:48.989 UTC	Coordinator connector commit.acknowledged `portal-20261103232137992-1:3671`

#3672
Raw inputs
	2026-11-03 23:22:48.993 UTC
	source 2026-11-03 23:22:48.970 UTC
		Raw input image, commit sequence 542
		`cycle 36132 · boot 1787886077538`
		▶ 20 values
		`portal-20261103232137992-1:3672`

#3673 PLC ack	2026-11-03 23:22:48.997 UTC	PLC applied commit 542 (45.4 ms) `portal-20261103232137992-1:3673`

#3674
PLC output
	2026-11-03 23:22:49.000 UTC
	source 2026-11-03 23:22:48.970 UTC
		PLC outputs at cycle 36132
		`cycle 36132 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:3674`

#3722 HMI action	2026-11-03 23:22:49.608 UTC source 2026-11-03 23:22:49.594 UTC	HMI action cmd.reset · PLC applied `portal-20261103232137992-1:3722`

#3731
PLC output
	2026-11-03 23:22:49.664 UTC
	source 2026-11-03 23:22:49.640 UTC
		PLC outputs at cycle 36165
		`cycle 36165 · boot 1787886077538`
		▶ 40 values
		`portal-20261103232137992-1:3731`

▶ Evidence #3733 – #6885 (27 records)

Printed copies include the first evidence group; open the HTML report for the remaining groups.

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